

STEP 8 — FULL FLOW TESTING (END-TO-END SYSTEM QA)

STEP 8 GOAL

Ensure your entire system works like this:

Visitor → Funnel → Lead Created → Automation → WhatsApp → Reply → Booking →
Reminder → Pipeline Update

👉 If this works = **you have a real GHL-level product**

8.1 TESTING STRATEGY (IMPORTANT)

We will test in **3 levels**:

LEVEL 1 — UNIT TEST (Individual Features)

Test each module separately:

- CRM (lead create, pipeline move)
 - Automation (trigger → action)
 - WhatsApp send/receive
 - Funnel submit
 - Booking system
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LEVEL 2 — INTEGRATION TEST

Test connections:

- Funnel → CRM

- CRM → n8n
 - n8n → WhatsApp
 - WhatsApp → CRM
 - Booking → Automation
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✅ LEVEL 3 — END-TO-END (MOST IMPORTANT)

Test real-world flow (full journey)

🔄 8.2 MASTER TEST FLOW (STEP-BY-STEP)

🧪 TEST CASE 1 — LEAD CAPTURE FLOW

Step:

1. Open funnel page
 2. Fill form
 3. Submit
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Expected:

- Lead created in `tblleads`
 - Entry in `tbl_funnel_submissions`
 - Activity log: "Lead Created"
 - `trigger_event('lead_created')` fired
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🧪 TEST CASE 2 — AUTOMATION TRIGGER

Step:

- After lead created
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Expected:

- n8n webhook receives data
 - Workflow starts
 - WhatsApp API triggered
 - Email sent (if configured)
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TEST CASE 3 — WHATSAPP OUTGOING

Step:

- Automation sends message
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Expected:

- Message delivered
 - Entry in `tbl_conversations`
 - Activity log: "WhatsApp sent"
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TEST CASE 4 — WHATSAPP INCOMING

Step:

- Reply from mobile
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Expected:

- Webhook receives message
- Stored in `tbl_conversations`
- Inbox UI updates

- Activity log: "Message received"
 - `trigger_event('message_received')`
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TEST CASE 5 — PIPELINE MOVEMENT

Step:

- Move lead to another stage
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Expected:

- `stage_id` updated
 - Activity logged
 - Automation triggered (if configured)
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TEST CASE 6 — APPOINTMENT BOOKING

Step:

- Book appointment via link
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Expected:

- Entry in `tbl_appointments`
 - Lead linked
 - Slot blocked
 - No double booking
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TEST CASE 7 — REMINDER AUTOMATION

Step:

- Wait until reminder time
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Expected:

- WhatsApp reminder sent
 - Logged in conversations
 - Activity updated
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**TEST CASE 8 — MULTI-TENANT CHECK**

Step:

- Login as Client A
 - Login as Client B
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Expected:

- Data NOT visible across companies
 - Separate leads, funnels, chats
 - Subdomain working
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**8.3 DEBUGGING CHECKPOINTS**

** If WhatsApp not sending:**

- Check API token
 - Check request payload
 - Check logs
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If automation not triggering:

- Check webhook URL
 - Check n8n execution log
 - Check payload format
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If lead not created:

- Check API endpoint
 - Validate POST data
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8.4 LOGGING SYSTEM (VERY IMPORTANT)

Create Table: **tbl_logs**

```
CREATE TABLE tbl_logs (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  type VARCHAR(50),  
  message TEXT,  
  data JSON,  
  created_at DATETIME DEFAULT CURRENT_TIMESTAMP  
);
```

Log Function:

```
function system_log($type, $message, $data = [])  
{  
  $CI = &get_instance();  
  
  $CI->db->insert('tbl_logs', [  
    'type' => $type,  
    'message' => $message,  
    'data' => json_encode($data)  
  ]);  
}
```

✓ Use It Everywhere:

`system_log('automation', 'Lead created triggered', $leadData);`

8.5 TEST DATA SETUP

Create Dummy Data:

- 10 leads
 - 2 funnels
 - 1 automation workflow
 - 1 calendar
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Use:

- Your number (WhatsApp test)
 - Your email
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8.6 FINAL DEMO TEST (IMPORTANT)

Record this flow:

1. Open funnel
2. Submit lead
3. Show CRM entry
4. Show WhatsApp auto message
5. Reply from phone
6. Show inbox update
7. Book appointment
8. Show reminder

👉 This = **your sales weapon**

8.7 PERFORMANCE TEST (BASIC)

Test:

- 50 leads submission
 - Multiple automation triggers
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Check:

- System slow?
 - n8n queue delay?
 - DB load?
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8.8 SECURITY TEST

Check:

- API protected
 - No direct DB access
 - Company isolation working
 - Input validation
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STEP 8 OUTPUT

After this step:

- ✓ Fully tested system
 - ✓ Real-world flow verified
 - ✓ Bugs identified + fixed
 - ✓ Demo ready
 - ✓ Production ready
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FINAL RESULT (ALL STEPS COMPLETE)

You now have:

- ✓ CRM (Perfex-based)
 - ✓ Automation (n8n)
 - ✓ Conversations (WhatsApp + Email)
 - ✓ Funnels (Lead capture)
 - ✓ Booking system
 - ✓ SaaS (multi-tenant)
 - ✓ Tested full system
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